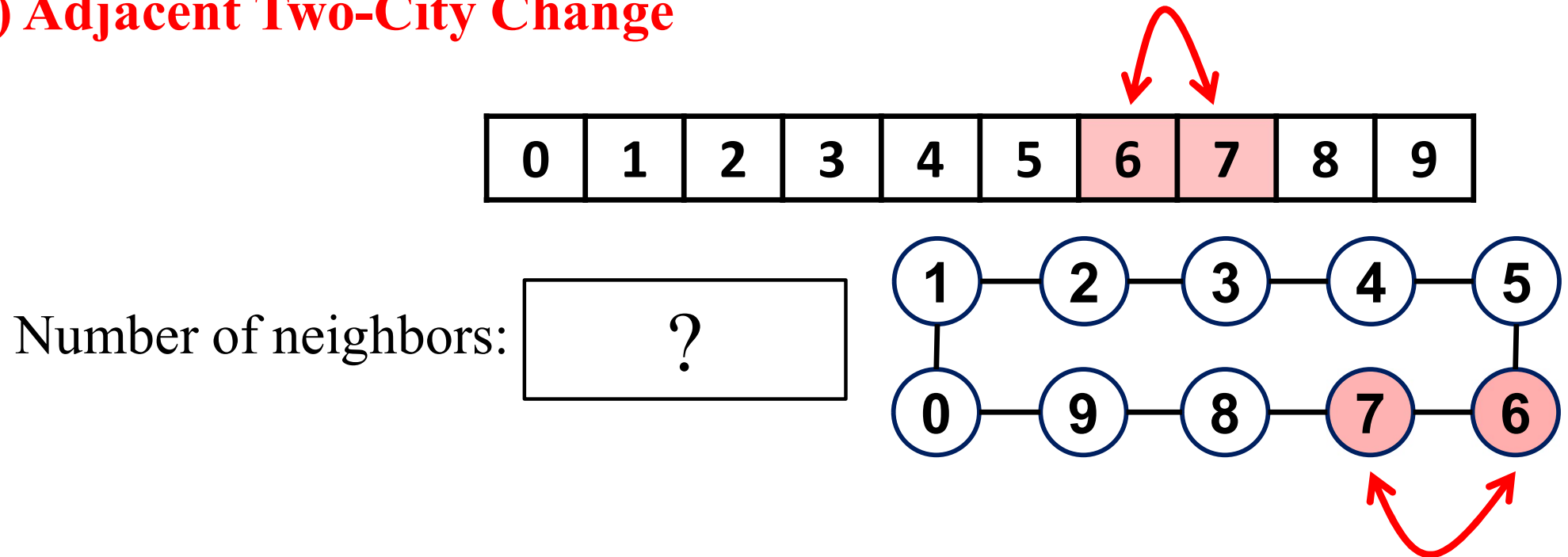


Lab Session Task:

Show the number of neighbors created by each of the following neighborhood structures for an n -city TSP problem.

(1) Adjacent Two-City Change

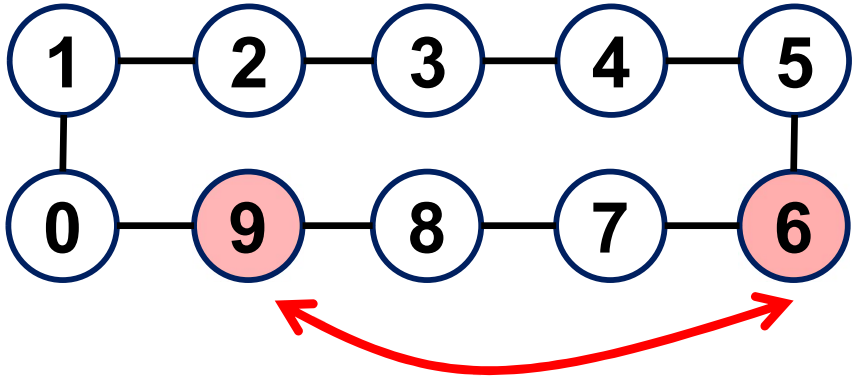
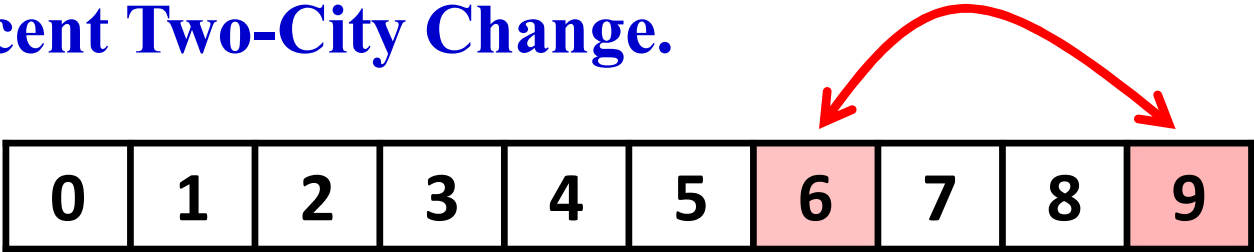


Your answer should be a formulation including “ n ” such as $n(n - 1)/2$ instead of a number such as 10 and 45.

(2) Arbitrary Two-City Change
including (1) Adjacent Two-City Change.

Number of neighbors:

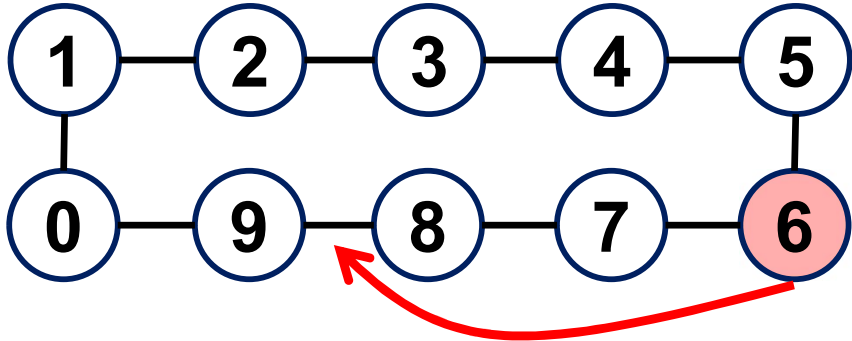
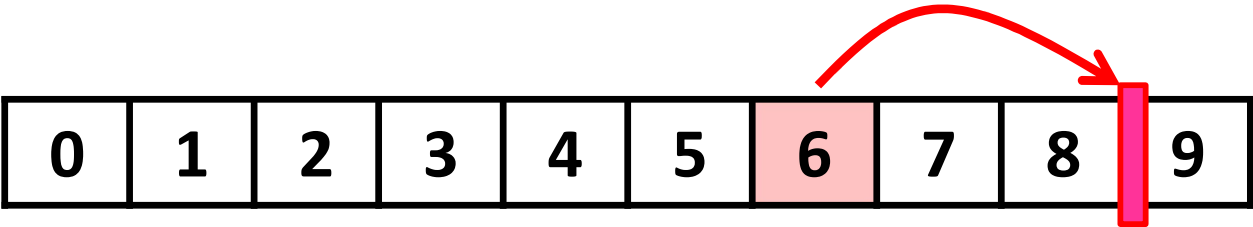
?



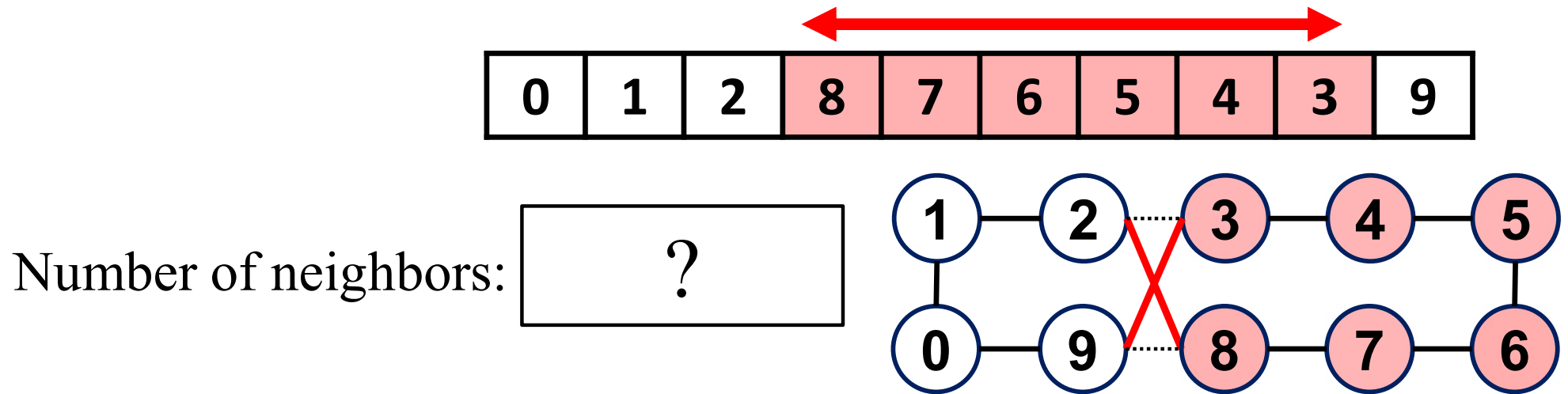
(3) Insertion (Shift)

Number of neighbors:

?



(4) Inversion (Arbitrary Two-Edge Change)



(5) Arbitrary Three-City Change including two-city change in (1) and (2).

